Marathwada Shikshan Prasarak Mandal's
Deogiri Institute of Engineering and Management Studies, Chhatrapati Sambhajinagar

Project Report on

Scholarship Navigator – Smart Student Funding Assistant

Submitted By

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Department of Computer Science and Engineering
Deogiri Institute of Engineering and Management Studies,
Chhatrapati Sambhajinagar
(2026- 2027)

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In partial fulfillment of

Bachelor of Technology

(Computer Science and Engineering)

Guided By

Mrs .Sugandha Nandedkar

Department of Computer Science and Engineering

Deogiri Institute of Engineering and Management Studies,

Chhatrapati Sambhajinagar

(2026- 2027)

CERTIFICATE

This is to certify that, the Project entitled "**Scholarship Navigator – Smart Student Funding Assistant**" submitted by **Akanksha Popat Jadhav (CS4101), Vaishnavi Rajendra Thombre (CS4102), Vaishnavi Sanjay Untwal (CS4143)** is a bonafide work completed under my supervision and guidance in partial fulfillment for award of Bachelor of Technology (Computer Science and Engineering) Degree of Dr. Babasaheb Ambedkar Technological University, Lonere.

Date:
Place: Chhatrapati Sambhajinagar

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Head of Department	Director
Dr. R. M. Autee	**Dr. Subhash. V. Lahane**

DECLARATION

This is to certify that, the Project Report entitled (TNRB 12), "**Scholarship Navigator – Smart Student Funding Assistant**" Submitted by **Akanksha Popat Jadhav (CS4101), Vaishnavi Rajendra Thombre (CS4102), Vaishnavi Sanjay Untwal (CS4143)** is a bonafide work completed under my supervision and guidance in partial fulfillment for award of bachelor's degree in Computer Science and Engineering of Deogiri Institute of Engineering and Management Studies, Chhatrapati Sambhajinagar under Dr. Babasaheb Ambedkar Technological University, Lonere.

Place: Chhatrapati Sambhajinagar
Date:

Mrs. Sugandha Nandedkar
Guide

External Examiner

Abstract (TNR-16/Bold/Centre)

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Justified

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CHAPTER 1: INTRODUCTION

Higher education can create important opportunities for personal and professional development, but tuition, accommodation, books, travel, and other academic expenses may create a financial burden. Scholarships and financial-aid programmes help reduce this burden. However, information about these opportunities is distributed across government portals, state portals, institutional websites, private foundations, and other online sources. Providers publish different eligibility conditions, application procedures, required documents, and deadlines. Students must therefore search several websites and compare their circumstances with multiple sets of requirements.

ScholarNexus is a web-based scholarship discovery and assistance platform intended to organize scholarship information and help students explore opportunities relevant to their educational profile. Depending on the modules implemented in the submitted build, it may support browsing, search and filtering, preliminary eligibility matching, and chatbot-based guidance. It assists discovery; it does not replace official scholarship rules or the final decision of a provider.

1.1 Existing System

In the existing environment, scholarship information is accessed through separate government portals, state portals, college notices, university websites, non-profit organizations, and private scholarship pages. These sources are useful, but their structure, terminology, update frequency, and level of detail vary.

- Distributed information: students may need to visit multiple websites to discover opportunities from different providers.
- Manual eligibility checking: students compare course, year of study, domicile, income range, category, academic performance, institution type, and scheme-specific conditions.
- Inconsistent presentation: award values and deadlines may appear on a webpage, notice, PDF, or lengthy eligibility text.
- Repeated effort: similar checks must be repeated across different schemes.
- Limited personalization: general listings may show many scholarships without indicating which appear relevant to a particular student.
- Information freshness: award values, eligibility rules, application windows, and links can change, making old listings misleading.

Some existing portals already provide search, filters, eligibility details, or personalized services. The limitation is not that every existing portal lacks these features; rather, students may still need to navigate separate sources and interpret the information independently. ScholarNexus aims to make discovery and comparison easier in one interface**.**

1.2 Problems or Need for Implementation

A centralized scholarship discovery system is needed because finding financial assistance is often fragmented and time-consuming. A student may qualify for several schemes but not know their names, where to find official information, or how to interpret the conditions. A missed deadline or misunderstood criterion can prevent an otherwise suitable application.

The system should reduce search effort while presenting scholarship details consistently. Students should be able to locate opportunities, inspect conditions, and proceed to official application sources. Missing award values or deadlines must not be invented, and a recommendation must not be treated as a guarantee of eligibility.

The project also provides practical experience with frontend-backend integration, database operations, collection of public information, user-oriented search, eligibility-matching concepts, and chatbot integration.

1.3 Proposed System

ScholarNexus proposes a web-based platform that organizes scholarship information and supports students during discovery. The logical workflow is to collect scholarship records from configured sources or managed entries, store available fields, and display them through a user interface. Students can browse records and use available filters or profile-based matching where those modules are implemented.

- User interface: pages for browsing scholarship records and accessing system features.
- Scholarship data management: records may include scheme name, provider, description, eligibility, award value, deadline, official URL, and other available fields.
- Data collection and processing: obtains information from permitted public sources and converts it into structured records, respecting source terms and access restrictions.
- Search and filtering: narrows results by available fields such as education level, course, state, category, or application status.
- Eligibility assistance: compares student-profile attributes with recorded scholarship conditions and distinguishes likely matches from uncertain cases.
- Chatbot assistance: accepts natural-language questions and returns guidance through the backend chatbot service.
- Administration and maintenance: supports reviewing, correcting, and refreshing records where these functions exist.

- The current codebase uses a React frontend and Flask backend, and the chatbot route is registered under /api/chatbot. The final report should describe only modules working in the submitted build. A proposed feature should not be called completed unless implemented and tested.

1.4 Objectives

The objectives are divided into learning objectives and system-development objectives to document both what was studied and what the project aims to deliver.

1.4.1 Learning and Study Objectives

- To study React.js, Flask, MongoDB, and API integration for web application development.
- To study web scraping and data processing techniques for collecting scholarship information.
- To study eligibility matching and machine learning techniques for scholarship recommendations.
- To understand chatbot integration, software testing, data security, and privacy.

1.4.2 Personal Learning Objectives

- To apply engineering knowledge to solve a real-world problem.
- To improve technical, analytical, and problem-solving skills.
- To gain practical experience in full-stack development and machine learning.

1.4.3 Student Benefits

- To reduce the time and effort required to search for scholarships.
- To help students discover suitable financial assistance opportunities.
- To improve awareness of scholarship eligibility criteria and deadlines.

1.4.4 System Objectives

- To develop a centralized platform for scholarship information from multiple sources.
- To provide scholarship search, filtering, and eligibility-checking facilities.
- To recommend suitable scholarships based on students' profiles.
- To provide scholarship details, official links, and chatbot assistance.In the final submission, mark

1.5 Problem Statement

To design and develop a centralized, user-friendly web-based Scholarship Navigator that collects and organizes scholarship information from multiple sources, enables students to search and filter

scholarships based on their personal and academic profiles, checks preliminary eligibility, provides personalized recommendations, and helps students track application deadlines through relevant notifications and official links.

1.6 Scope of the Project

1.6.1 In Scope

- To develop a web-based platform for browsing and managing scholarship information.
- To collect and organize scholarship details from selected sources.
- To provide search and filtering based on students' academic and personal criteria.
- To support preliminary eligibility checking and suitable scholarship recommendations.
- To integrate a chatbot for scholarship-related queries.
- To validate scholarship data and display available deadlines and official application links.

1.6.2 Out of Scope and Limitations

- The system does not award scholarships or guarantee funding.
- Eligibility results are preliminary and must be confirmed by the official scholarship provider.
- The system may not cover all available scholarships.
- Scraped information may be incomplete or outdated.
- Automated application submission and advanced real-time notifications are outside the scope unless implemented and tested.
- Student information must be handled securely and with appropriate privacy protection**.**

The scope prioritizes organized discovery and assistance rather than replacing official scholarship portals or making financial-aid decisions.

1.7 Organization of the Report

Chapter 1 introduces the existing system, need for implementation, proposed system, learning and system objectives, problem statement, scope, and report structure. Chapter 2 reviews research on scholarship eligibility expert systems, educational recommendation, course recommendation, fairness, and explainability; compares the studies critically; and identifies the research gap. If the complete report contains additional chapters, they may describe system design, implementation, testing and results, conclusion, and future enhancements according to the institution's format.

CHAPTER 2: LITERATURE SURVEY

A literature survey examines prior research related to a project, identifying methods already explored, their contributions, limitations, and remaining gaps. For ScholarNexus, relevant areas include scholarship eligibility expert systems, educational recommender systems, course recommendation, data quality, and responsible recommendations.

Scholarship discovery combines information retrieval and decision support. A student provides profile details while each scholarship has a set of conditions. The system retrieves relevant records, compares known profile attributes with conditions, and presents results for verification. Course recommendations cannot automatically be treated as scholarship eligibility because scholarships may contain mandatory conditions, financial-need criteria, regional rules, documents, and deadlines.

2.1 Introduction to the Literature Survey

Three approaches are relevant. Rule-based expert systems encode eligibility conditions as explicit rules. Recommender systems rank or filter items based on profiles, item attributes, or interaction patterns. Data collection and information extraction convert public information into structured records. Rules may become outdated, recommenders may be opaque or biased, and extracted data may be incomplete or incorrect; therefore these methods need validation.

2.2 Review of Important Research Papers

2.2.1 Research Paper: Rule-Based Expert Systems for Supporting University Students

Engin et al. (2014) presented a research paper titled Rule-Based Expert Systems for Supporting University Students, published in Procedia Computer Science, Volume 31, pages 22–31. The paper describes an expert system developed to recommend scholarships to university students based on their eligibility criteria, such as academic performance, income, and other requirements. The system used Oracle Policy Automation to apply predefined rules and identify suitable scholarships. It also explained why a student was eligible or ineligible for a particular scholarship. However, collecting complete scholarship information and maintaining updated eligibility rules were major challenges. This research provides a foundation for developing a scholarship recommendation system that helps students identify suitable financial assistance opportunities.

2.2.2 Existing Project: Scholarship Mitra

Scholarship Mitra is an existing scholarship portal developed using React, Flask, and MySQL. It provides scholarship information and helps students find opportunities based on eligibility criteria. The project includes features such as student registration, scholarship searching, eligibility filtering,

recommendation ranking, and an administrative dashboard. Its recommendation approach uses eligibility rules and similarity-based ranking to identify relevant scholarships. However, its usefulness depends on the completeness and accuracy of the stored scholarship data. The Scholarship Navigator project aims to build on this concept by using MongoDB Atlas and incorporating centralized scholarship information, personalized matching, and deadline-related assistance.

2.2.3 Existing Project: AI-Driven Scholarship Finder

AI-Driven Scholarship Finder is a software project designed to provide personalized scholarship recommendations using rule-based filtering and AI-based semantic matching. According to its repository, it uses React and Flask, applies eligibility rules, and ranks scholarship descriptions according to their relevance to student profiles. It also includes a chatbot and email notifications to help students understand scholarship opportunities and deadlines. The project demonstrates how traditional eligibility filtering can be combined with AI-based recommendation techniques. However, the reliability of its recommendations depends on the quality of the scholarship dataset, the accuracy of extracted eligibility conditions, and the freshness of deadline information.

2.2.4 Existing Project: Scholarship Recommendation System

The Scholarship Recommendation System is another software-based approach that uses student information and scholarship eligibility criteria to suggest relevant opportunities. Such systems commonly combine rule-based eligibility checking with text-similarity techniques, such as TF-IDF, to rank scholarships according to the student's profile. These techniques can reduce manual searching and help students identify relevant scholarships. However, recommendation quality depends on the dataset and the criteria used by the system. The Scholarship Navigator project focuses on bringing scholarship information from selected sources into one platform and providing search, eligibility matching, and deadline-related assistance.

2.3 Critical Analysis and Comparative Evaluation

No single method solves every aspect of scholarship discovery. Rule-based systems suit explicit eligibility conditions; content-based methods use item attributes and stated preferences; collaborative methods require meaningful interaction data; hybrid methods combine signals but increase complexity. Data quality is foundational: a sophisticated model cannot compensate for a false award amount, outdated deadline, or incorrectly extracted condition unless the data pipeline detects and corrects the error.

The following table compares the research paper and existing projects with the proposed Scholarship Navigator – Smart Student Funding Assistant.

No.	Research paper / Existing project	Approach / Technology	Main contribution	Critical limitation
1	Rule-Based Expert Systems for Supporting University Students (2014)	Rule-based expert system using Oracle Policy Automation	Checks scholarship eligibility and explains the results.	Requires scholarship rules and information to be maintained and updated.
2	Scholarship Mitra	React, Flask, MySQL, eligibility rules, similarity-based ranking	Provides scholarship searching, eligibility filtering, recommendations, and an admin dashboard.	Depends on the quality and completeness of the stored scholarship data.
3	AI-Driven Scholarship Finder	React, Flask, rule-based filtering, semantic matching, chatbot, email alerts	Combines eligibility filtering with AI-based matching and notification features.	Incorrectly extracted criteria or outdated data may affect recommendation accuracy.
4	Scholarship Recommendation System	Eligibility rules and text-similarity techniques	Matches scholarship information with student profiles.	The quality of recommendations depends on the dataset and matching method.
5	Proposed Scholarship Navigator	React, Flask, MongoDB Atlas, and scikit-learn	Aims to centralize scholarship information and provide search, eligibility matching, personalized recommendations, and deadline assistance.	Data coverage, scraping reliability, deadline accuracy, and recommendation performance must be tested and maintained.

2.3.1 Key Findings from the Critical Analysis

- Do not label a scholarship as a confirmed match if a mandatory condition is unknown or contradicted.
- Treat freshness as part of data quality; award values, application windows, and eligibility

rules can change.

- Explain which criteria appear to match and which require confirmation.
- Choose methods that fit available data; collaborative filtering is not justified without sufficient interaction history.
- Test realistic profiles, missing fields, stale pages, and official-link access.
- Collect only necessary profile data and avoid unsupported assumptions.

2.4 Research Gap

The reviewed research and projects demonstrate that rule-based eligibility checking and AI-based recommendation can help students find relevant scholarships. However, challenges remain in collecting information from multiple sources, removing duplicate records, keeping deadlines updated, and presenting reliable eligibility information in one place.

The proposed Scholarship Navigator aims to address these challenges by providing a centralized platform for scholarship discovery, student-profile-based matching, filtering, and deadline assistance. Its effectiveness will depend on data validation, reliable source links, and testing the accuracy of eligibility and recommendation results. The system is intended to guide students toward suitable opportunities; final eligibility and application decisions remain subject to the official scholarship provider's rules.

Chapter 3: SYSTEM DEVELOPMENT

1. Use Case Diagram

The Use Case Diagram illustrates the interaction between the system's actors, such as Students and Administrators, and the various functionalities provided by the Scholarship Navigator system. It represents the functional requirements of the system from the users' perspective.

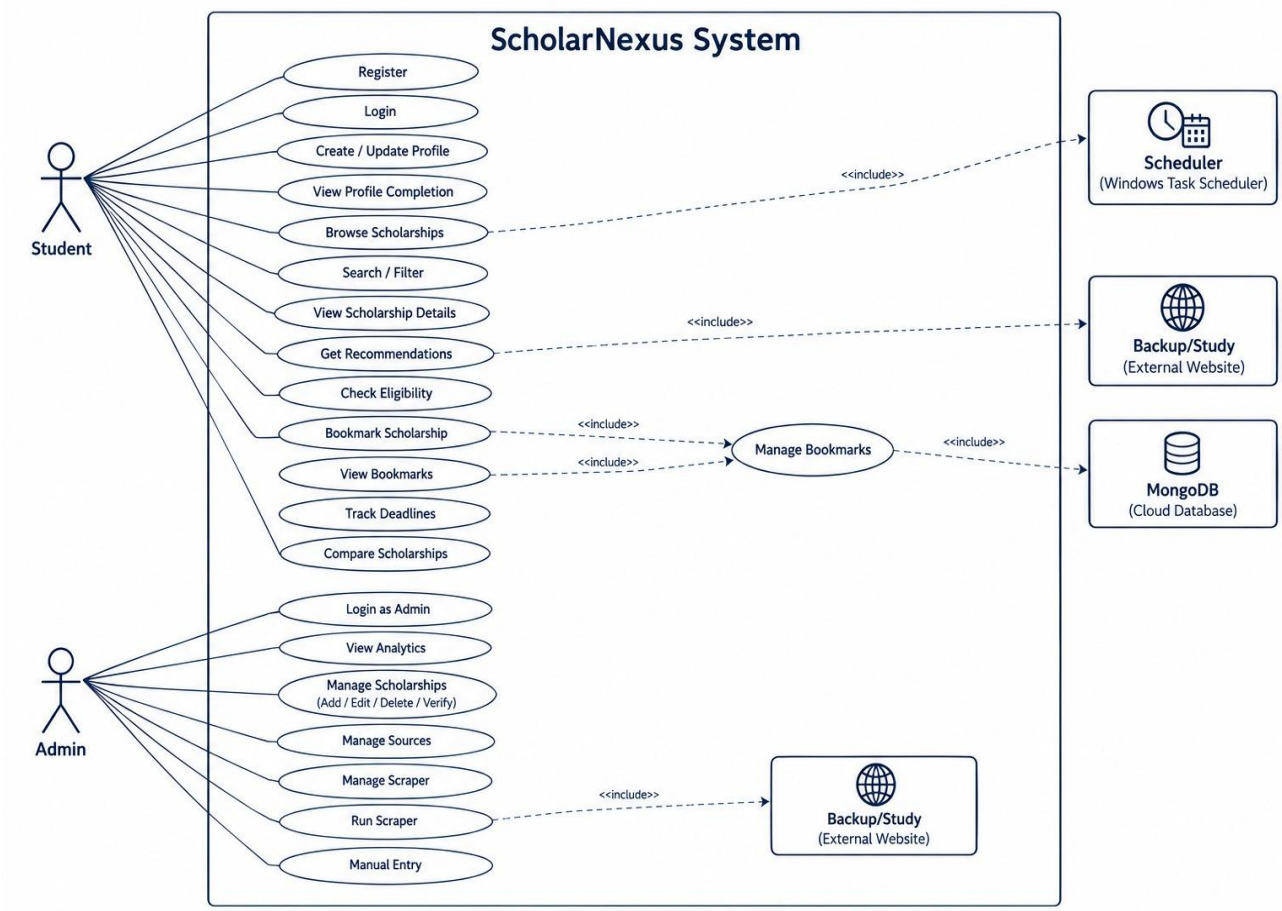


Figure 3.1 : use case diagram

2. DFD Level 0 (Context Diagram)

The Level 0 Data Flow Diagram (DFD) represents the Scholarship Navigator system as a single process and shows its interaction with external entities. It illustrates the flow of data between students, administrators, scholarship source we bsites, and the system

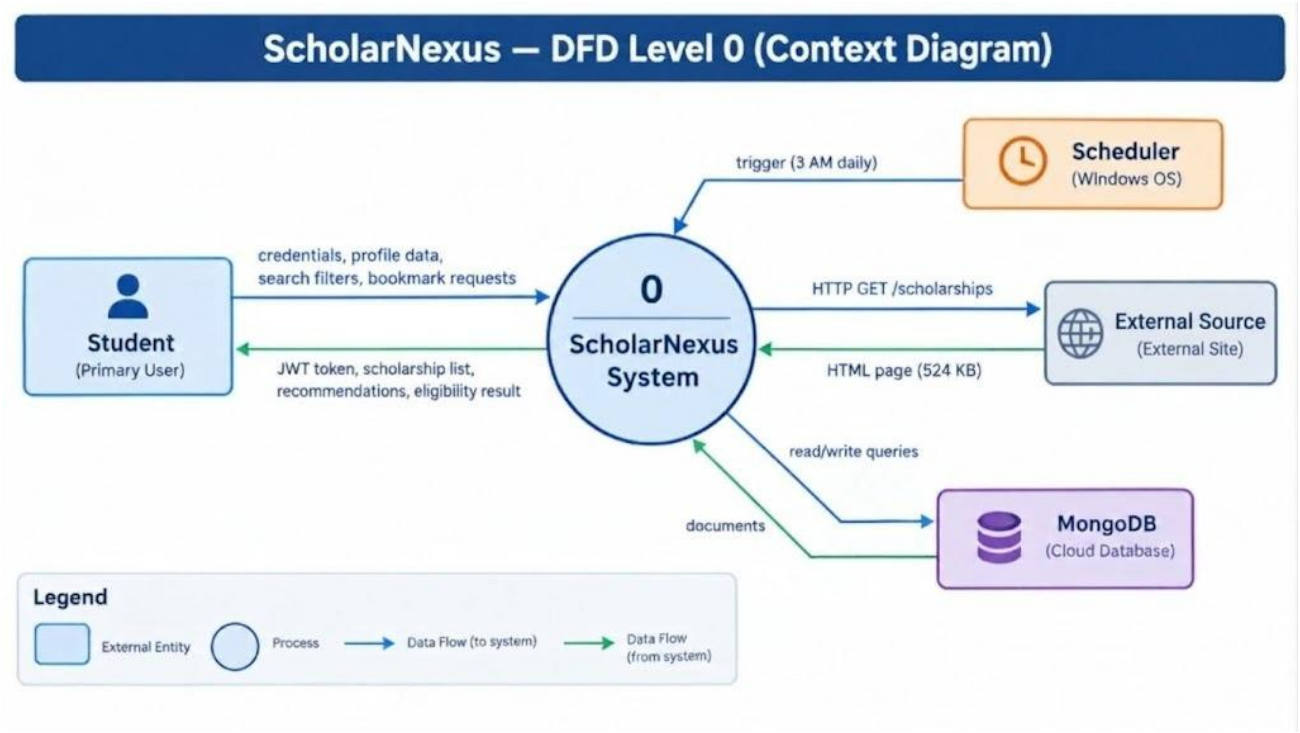


Figure 3.2 : DFD level 0

3. DFD Level 1

The Level 1 Data Flow Diagram provides a detailed representation of the major processes within the Scholarship Navigator system. It shows how scholarship data is collected, processed, stored, matched with student profiles, and presented as recommendations.

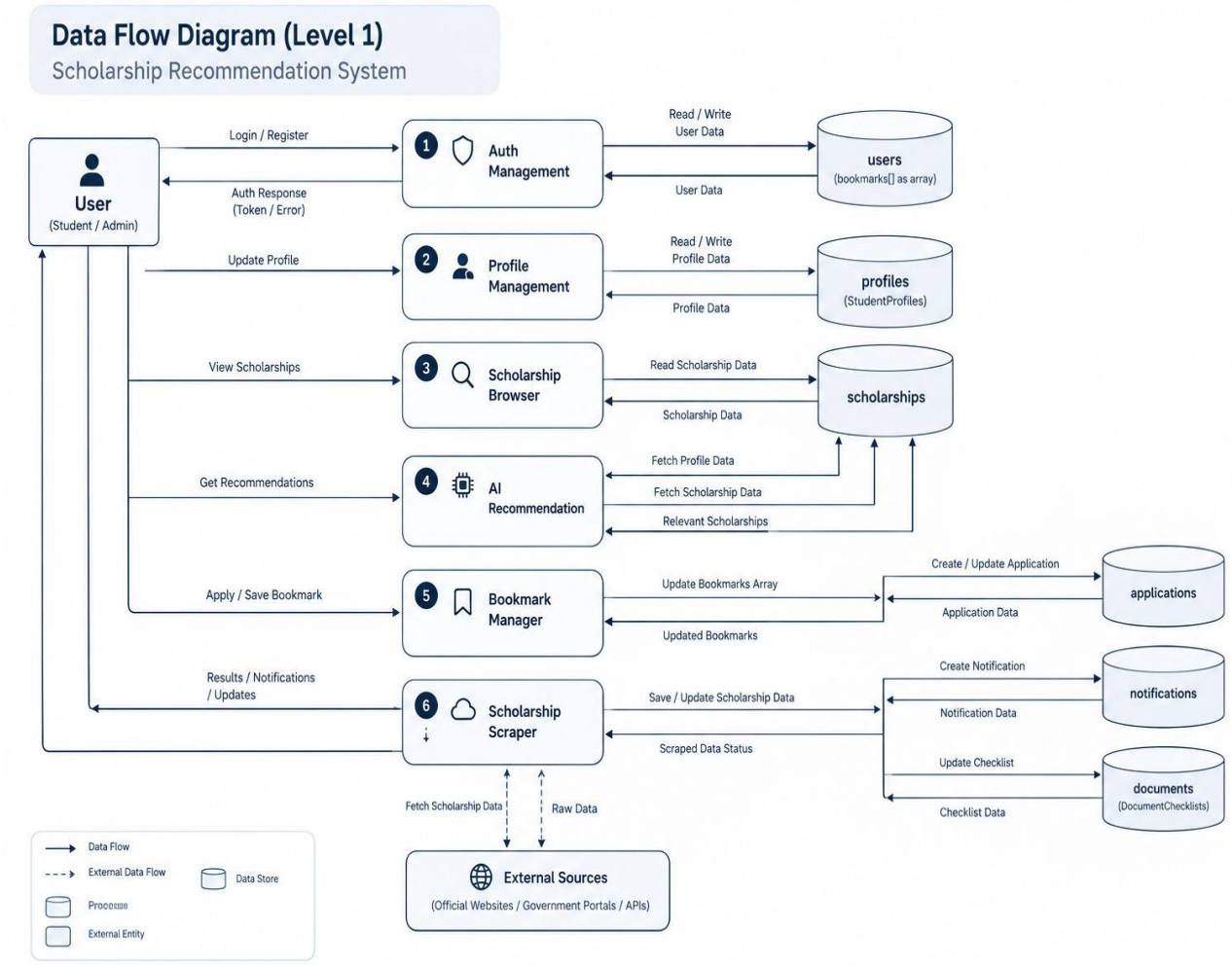


Figure 3.3 : DFD level 1

4. State Diagram

The State Diagram represents the different states of a scholarship record throughout its lifecycle, from discovery and data extraction to validation, publication, and archiving. It also illustrates the transitions between states based on system operations and administrative actions.

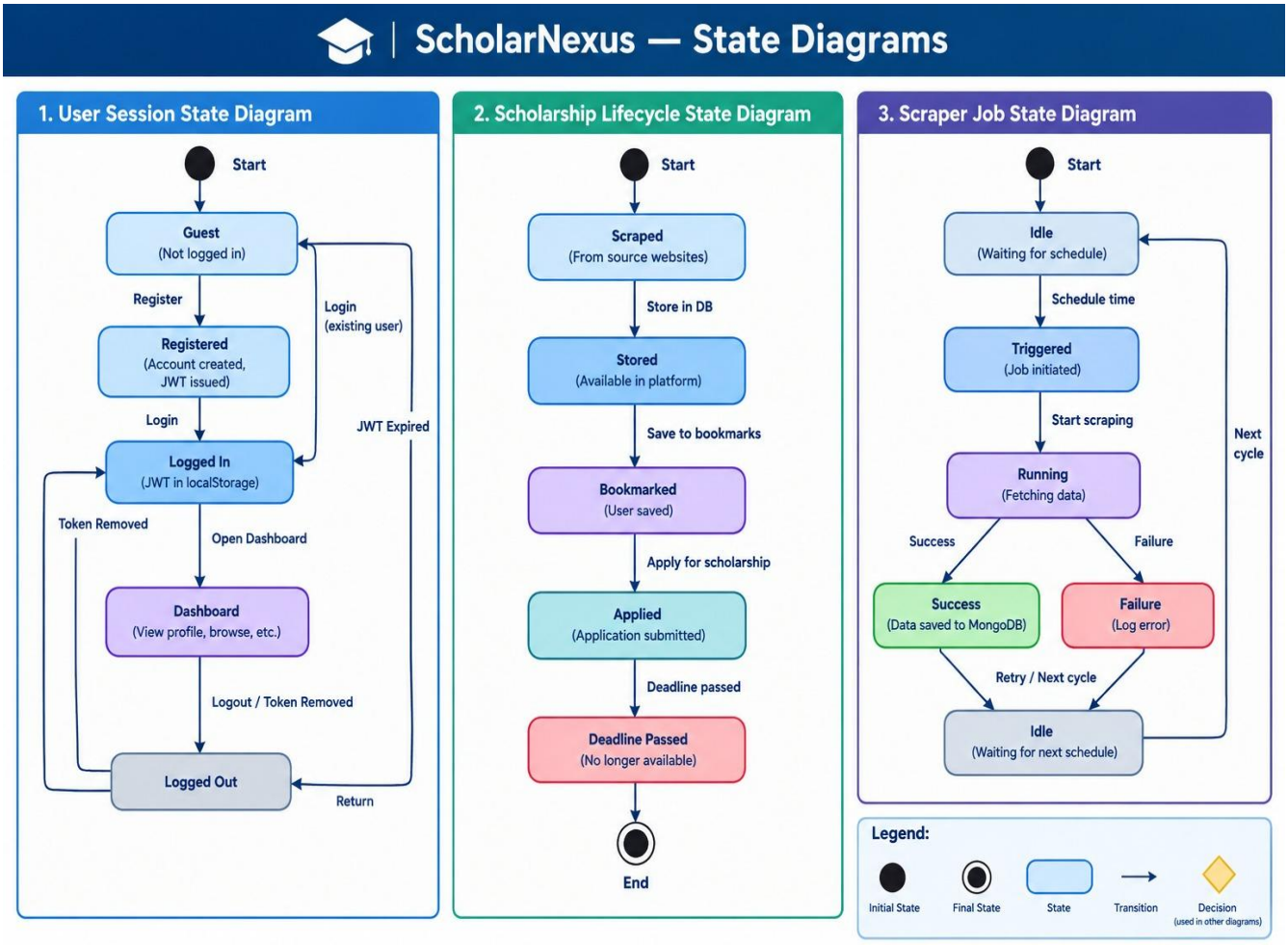


Figure 3.4: state diagram

5. Sequence Diagram

The Sequence Diagram illustrates the time-ordered interactions between the student, user interface, profile manager, matching engine, scholarship database, and recommendation ranker. It shows how these components communicate to generate and display suitable scholarship recommendations.

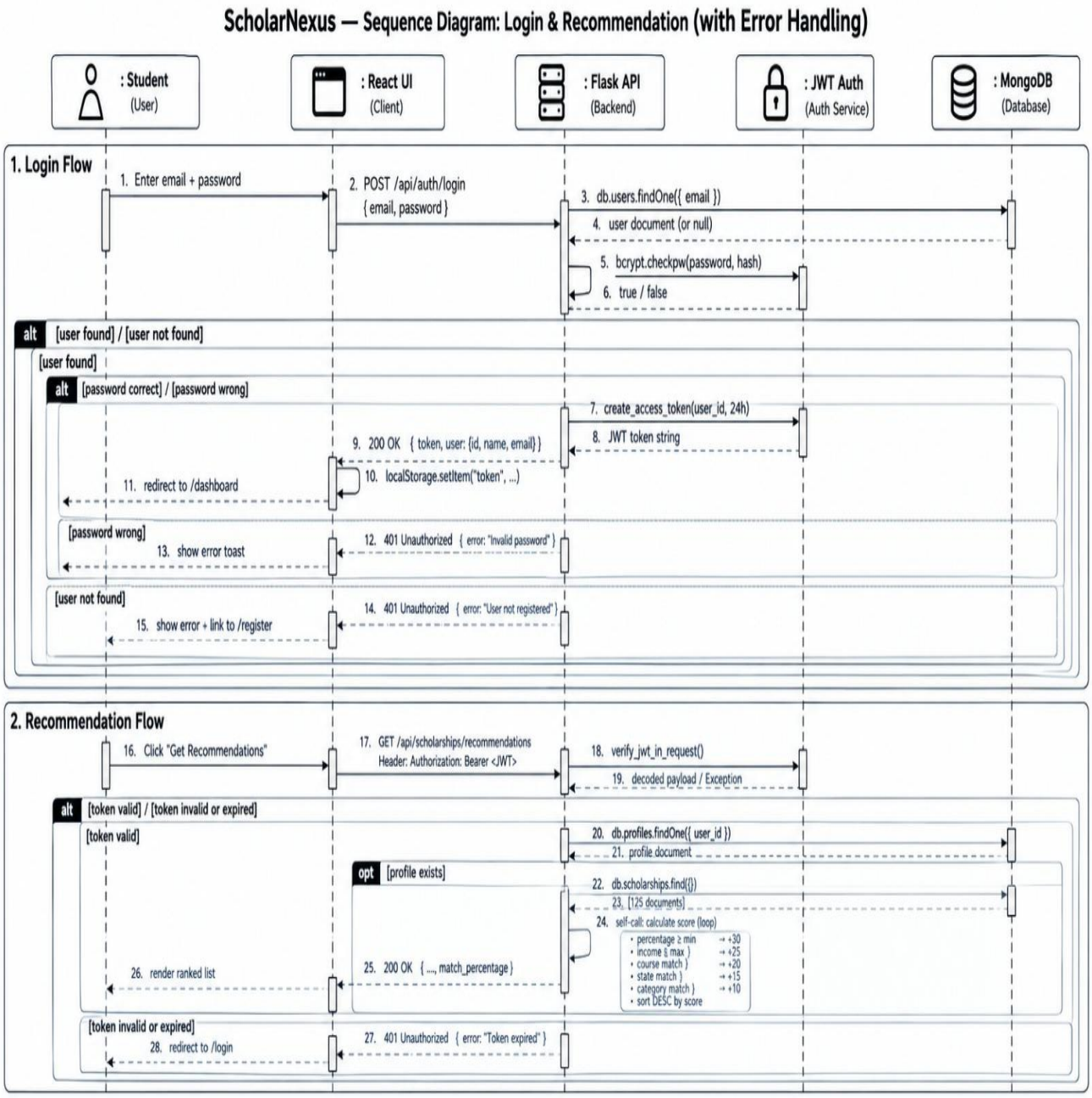


Figure 3.5: Sequence Diagram

6. Activity Diagram

The Activity Diagram represents the workflow of the Scholarship Navigator system, from collecting scholarship information and validating data to checking student eligibility and displaying recommendations. It also shows decision points that determine whether records are published, flagged for review, or filtered out.

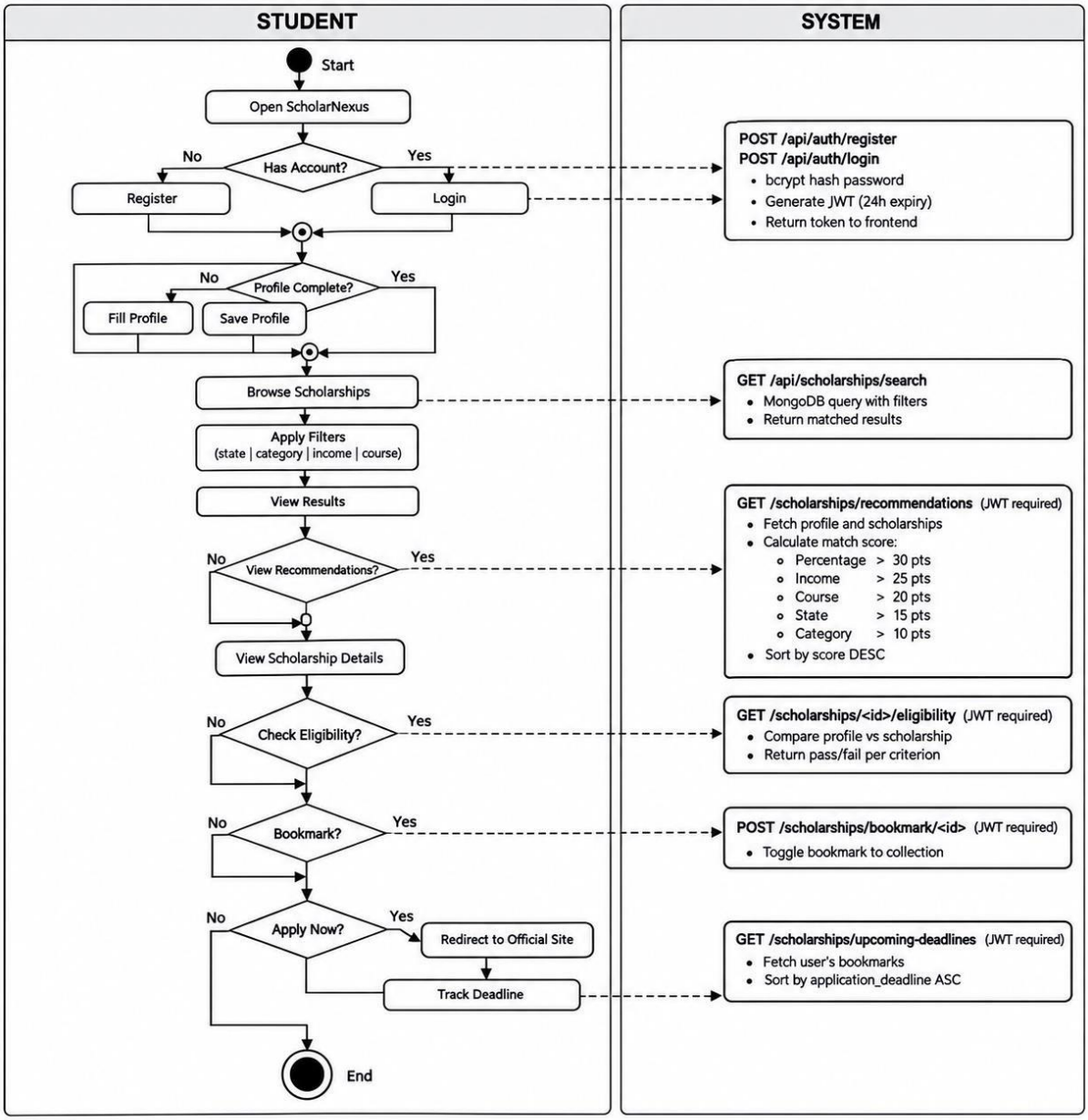


Figure 3.6: activity diagram

7. Database Design / ER Diagram

The Entity-Relationship (ER) Diagram represents the database structure of the Scholarship Navigator system, including entities such as Student, Scholarship, Source, Admin, and Match_Result. It illustrates the relationships between these entities and identifies their primary and foreign keys to support organized data storage and retrieval.

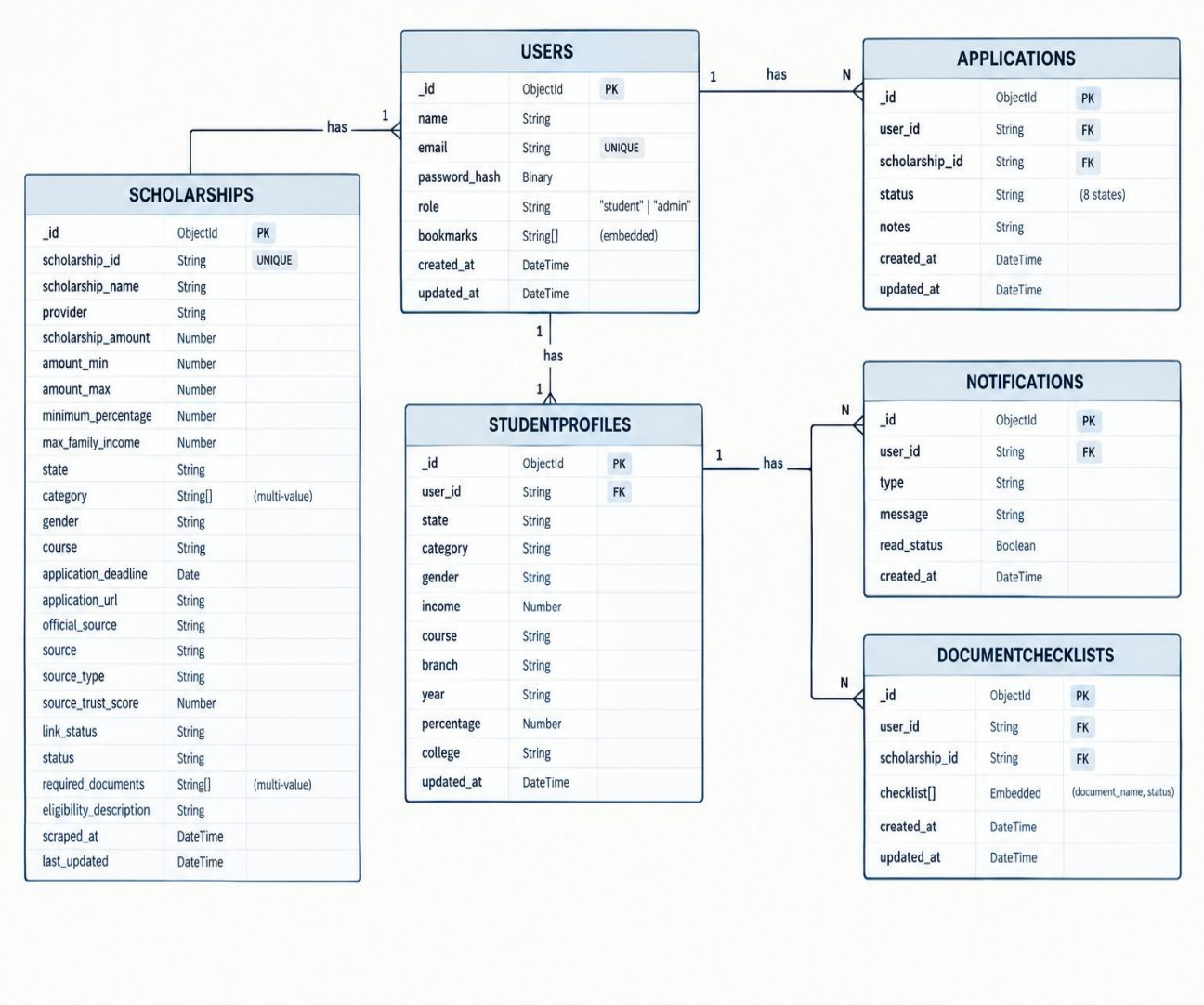


Figure 3.7 : database diagram

REFERENCES (TNR 16)(IEEE Format)

1. ***G. Engin, B. Aksoyer, M. Avdagic, D. Bozanlı, U. Hanay, D. Maden, and G. Ertek, "Rule-based expert systems for supporting university students," Procedia Computer Science, vol. 31, pp. 22–31, 2014. [Online]. Available: https://doi.org/10.1016/j.procs.2014.05.241. Accessed on: Oct. 16, 2026.***

2. **V. Poojari, "Scholarship Mitra—Smart Scholarship Recommendation & Management Portal," GitHub Repository. [Online]. Available: https://github.com/vighnesh116/scholarship-portal. Accessed on: Oct. 16, 2026.**

3. **B. Sreevardhan, "AI-Driven Scholarship Finder," GitHub Repository. [Online]. Available: https://github.com/bsreevardhan/AI-Driven-Scholarship-Finder. Accessed on: Oct. 16, 2026.**

4. **[4] A. Kammath, "Scholarship Recommendation System," GitHub Repository. [Online]. Available: https://github.com/Anjumkammath/scholarship-recommendation-system. Accessed on: Oct. 16, 2026.**

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Signature(s) of Students

Name of Students Sign
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- Chapter Text: 12 TNR
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- Page numbering should start from List of Figures page in roman (iii, iv etc.) and chapter page numbering from 1 onwards
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